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PROJECTS

KubeStash

Go • System Design • Clean Architecture • Backend • Kubernetes • Cloud • Concurrency • Prometheus • Grafana

A complete disaster recovery solution for Databases, Workload applications and Cluster for Kubernetes environment.

- Led design & development and ownership for 2.5 years from system design to release.
- Architect the full backend of the bucket as backup storage for all major clouds, with cloud identity integration, e.g., IRSA, Pod identity.
- Engineered high-performance backup & restore plugins (concurrent, snapshot, PITR) for 10+ databases.
- Added features (retry, validation, and failure recovery) to make it a leading k8s disaster recovery solution.
- Engineered high-performance backup & restore plugins (concurrent, snapshot, PITR) for 10+ databases.
- Architected sharded operators and controllers for large-scale, parallel backups across approx 1000+ DBaaS.
- Wrote Prometheus exporters and designed a Grafana monitoring dashboard.

KubeDB WAL-G

Go • Microservice • gRPC • Databases • Distributed System • Chunking

Wal Archival and Restoration for Databases in the Cloud

- Designed a microservice framework for database Point-in-Time Recovery (PITR) WAL archive.
- Implemented continuous WAL shipping to any object storage.
- Architected a gRPC proxy server for WAL streaming in Kubernetes distributed environments.
- Automated DBaaS failover detection and restore from WAL backup.

TopoLvm CSI Driver

Go • gRPC • FileSystems • LVM • Chunking • Restic • Prometheus • Grafana

Restic-based Volumes snapshot feature for kuberntes PVC

- Led the design & development of Volume Snapshot feature using concurrent chunking by Kubernetes CSI workflows.
- Efficient backup and restore operations for persistent volumes for hybrid and on-prem setups.
- Wrote Prometheus Prometheus exporters for monitoring volume snapshot operations.

Migrator

Go • gRPC • Protocol Buffers • Prometheus

High-performance data migration engine for PostgreSQL & MySQL

- Architect and Developed parallel/concurrent pipelines for high-volume database data transfer.
- Designed a gRPC-based control plane for managing complex migration workflows.
- Achieved approx +-10 hours to migrate 2 TB of MySQL data from RDS to kubedb managed DBaaS.

TaskQueue

Go • Concurrency • Parallel Processing

A Kubernetes controller that manages a queue of distributed task in kuberntes environment

- Built a Kubernetes controller for managing distributed task queues with concurrent processing capabilities.
- Implemented worker pool pattern for efficient task execution and resource management.

AWS Credential Manager

AWS • IAM • EC2 • EKS • Secret management • IRSA

A Credentials manager who managed permission of a Role Inside kubernetes

- Led the development of tools to manage credentials and permissions inside the Kubernetes cluster.

Go Shell

GO • Concurrency • Pipe • Buffering

A shell package to run multiple concurrent leaf commands using a single pipe input

TECHNICAL SKILLS

Languages: Go (Golang), C++, Bash

Platforms: Linux, Docker, Kubernetes, Openshift, Cloud-Native, AWS, GCP

Technologies: REST, gRPC, Protocol Buffers, Makefile, Helm, CI/CD, GitOps, FluxCD, Prometheus, Grafana

Storage systems: LVM, ZFS, K8s-CSI, Restic, Wal-G, Velero, Rook-Ceph, PostgreSQL, MySQL

Core Strengths: Microservice, Distributed Systems, Raft, Concurrency, Parallel Processing, Software Architecture